

AOS Controller Manual

Introduction

I. Current Version

- Manual version: v0.3.2
- Compatible AOS version: v0.3.2
- Compatible AOS Gamepad Server version: v0.3.2

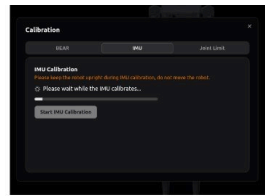
II. Document Description

The AOS Controller UI is designed to work with handheld consoles or mobile phones paired with a game controller. It provides a simple yet powerful interface for operating Westwood Robotics systems powered by AOS.

This document uses the THEMIS humanoid robot system running AOS as an example, though other robot systems follow similar operating methods.

Operation Instructions

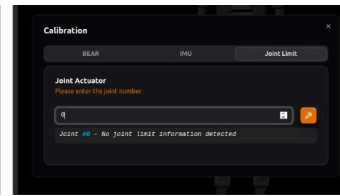
I. Overview of the Interaction Pages



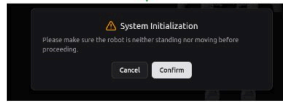
IMU Calibration



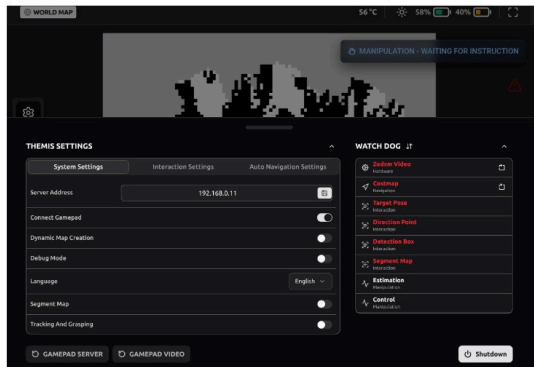
BEAR Calibration



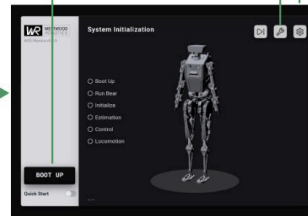
Joint Limits



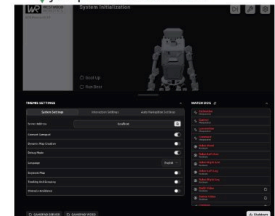
A prompt appears at the first startup step, reminding you to make sure the robot is not in a standing position.



System Settings Menu

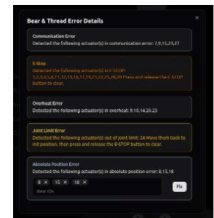
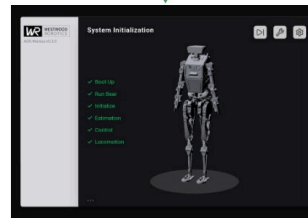


System Initialization Menu



System Settings Menu

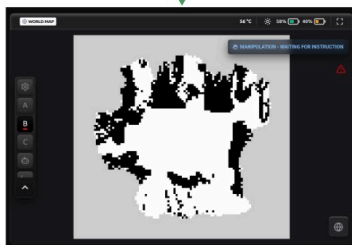
After tapping the **BOOT UP** button on the left, the system runs steps on the right in sequence. Once all steps are complete, the following is displayed:



Tapping to return to the initialization page.

The app then automatically navigates to the Main Operation Page — FPV:

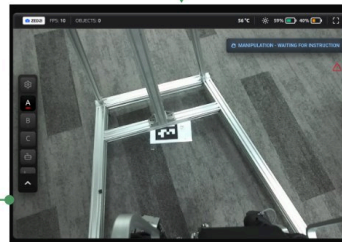
Tapping **B** in the vertical menu on the left navigates to the main operation page — World Map:



Main Operation Page - World Map

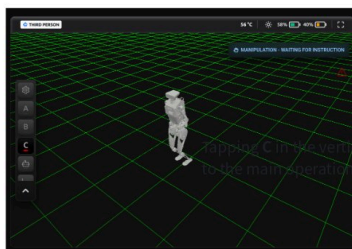


Navigation Bar



Main Operation Page — FPV

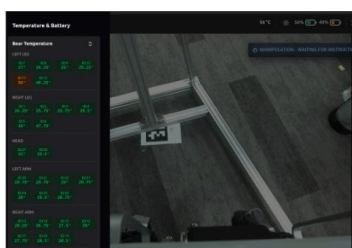
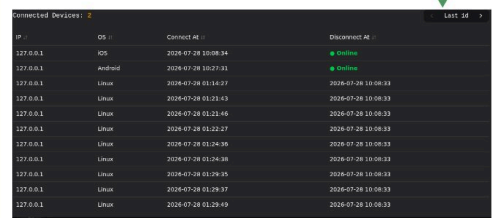
Tapping the flashing red triangle to open an Error Details dialog. Variants depending on use:



Main Operation Page - Third Person

Tapping **C** in the vertical menu on the left navigates to the main operation page — Third Person

Tap to view the connection information and IP address of the connected robot.



Tapping the line chart icon to open a panel of robot's motor temperatures and battery status:

Overview of the Interaction Pages

II. Functions of Individual Pages

1. Connecting to the Robot



Before running the Remote Control UI, please ensure that the remote control device is connected to the robot's onboard Wi-Fi.

Approximately 50 seconds after the robot is powered on, the onboard router will finish booting and begin broadcasting the onboard Wi-Fi SSID. The SSID naming convention is `RobotName_Band` (robot name + frequency band), for example `THEMIS_5G`. The default password is `gothemis`.

1.1 Open the Operation Page in a Browser

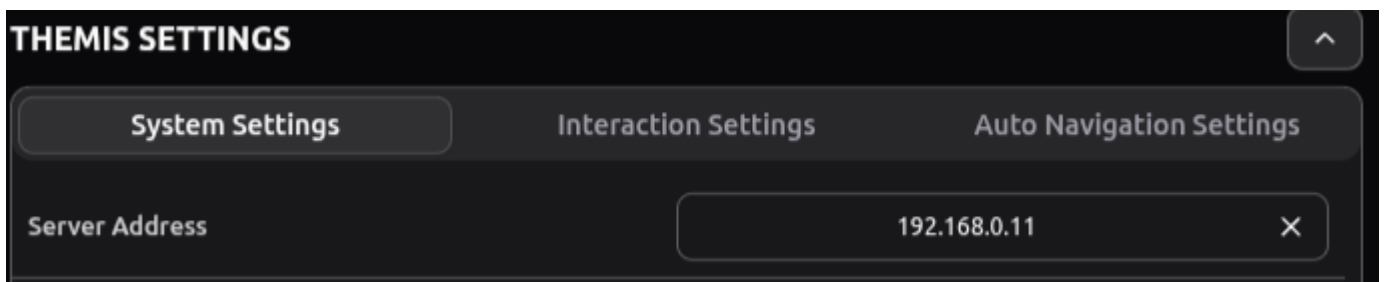
Accessing the Controller UI in a Web Browser, preferably Firefox or Chrome. In the address bar, enter `http://192.168.0.11:5000` (the default address) to open the login page. The default username and password are both `themis`. After a successful login, the System Settings page opens automatically. Please note:



Note: If you have modified the robot's router address range or the onboard computer's IP address, the UI address will change accordingly.

1.2 Establishing the Data Connection

- If the control device has already established a data connection with the robot, the black main button on the left is clickable. If it is grey, no connection has been established yet: first go to **System Settings** > enter the **Server Address** > Save, and wait until the connection succeeds before continuing.
- Click the gear icon in the upper-right corner, open the settings drawer at the bottom of the screen, and enter the robot IP (192.168.0.11 by default).

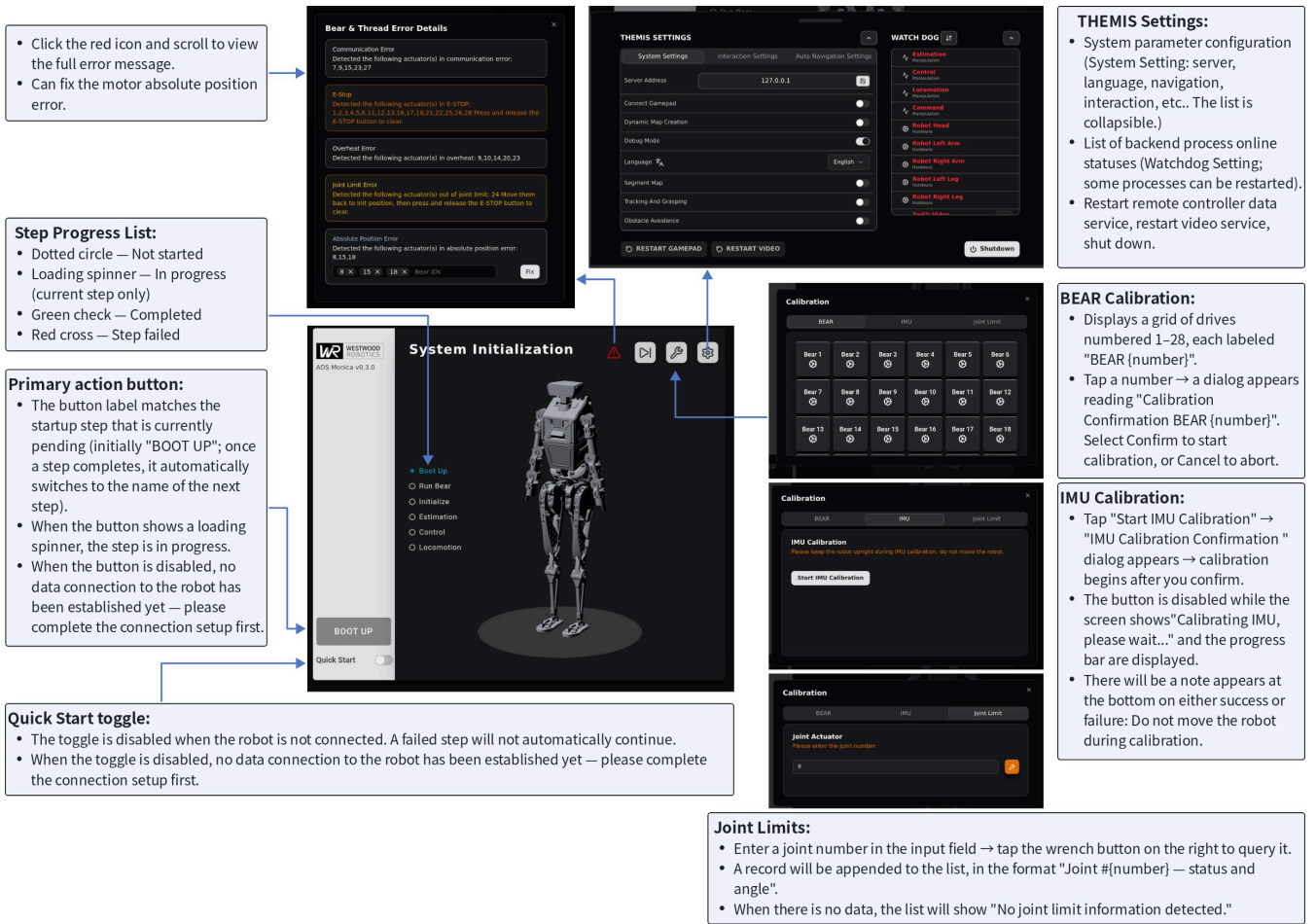


2. Initialization Page

- The initialization page is mainly used to start the subsystems of the robot in a fixed order (Boot Up -> Run BEAR -> Initialize -> Estimation -> Control -> Locomotion).
- The main button on the left executes the current step and the step list on the right shows the progress. The 3D robot model in the background changes in real time with joint states and body pose.
- After all steps have succeeded, the system automatically enters the main control interface.
- Other functions:

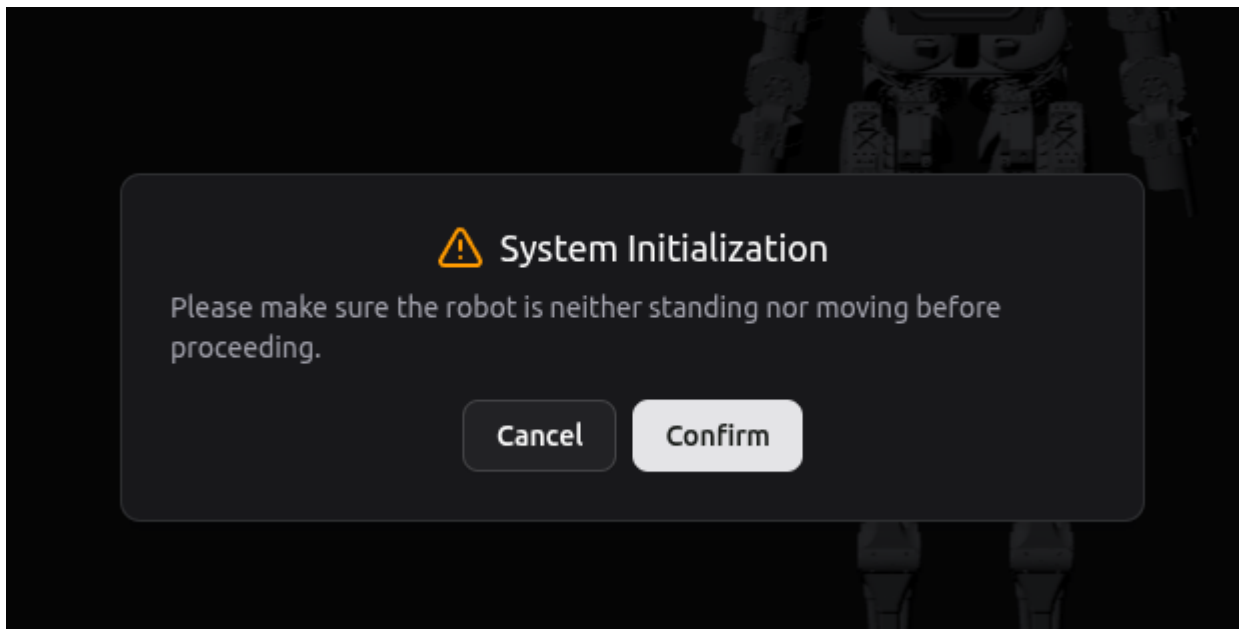
Module Location	Function	When to Use
Right background – robot model	Three-dimensional posture preview of the robot during the startup stage, updated with real-time joint / body data	Only for observing whether the posture is normal; it cannot replace the button on the left for completing the startup steps
Upper-right — red warning icon	Appears only when a BEAR driver fault or a thread status fault exists; click it to view the error type and the motor IDs involved. Some absolute-position errors can be diagnosed and, where supported, a repair can be attempted from the popup	Motor errors, motor overtemperature, thread errors, startup step errors, etc. The icon keeps flashing; If the icon is not displayed, none of the listed errors has been detected
Upper-right — wrench icon	Maintenance functions such as motor calibration, IMU calibration and joint limit query	Not part of the standard power-on procedure; use as needed when something has to be calibrated or configured
Upper-right — gear icon	Pops up the THEMIS Settings drawer menu from the bottom, including system parameters, process monitoring (Watchdog), restarting the data and video service, shutdown, etc.	When the connection fails, when the server IP or language has to be changed, or when a service has to be restarted
Upper-right — skip icon	Goes directly to the main control interface without executing the remaining robot startup steps	Use this button only when the remaining startup steps need to be skipped

- For detailed information on the interface elements, refer to the figure below:

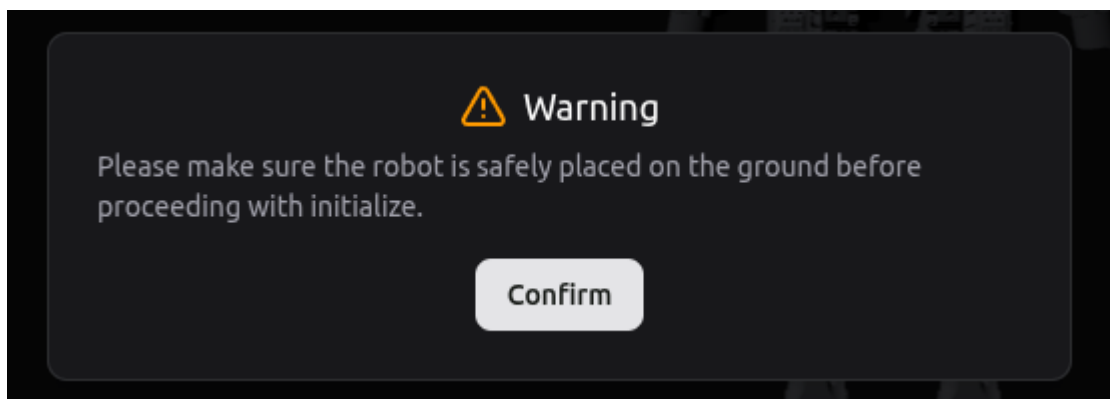


Initialization Page Functions

- !** When executing the first step "Boot Up", make sure that the robot is neither standing nor moving, otherwise the robot may fall over (this step shows a confirmation prompt as below).



⚠ Before executing the “Initialize” step, the robot must be placed safely and stably on the ground (this step shows a confirmation prompt as below).



3. FPV Page

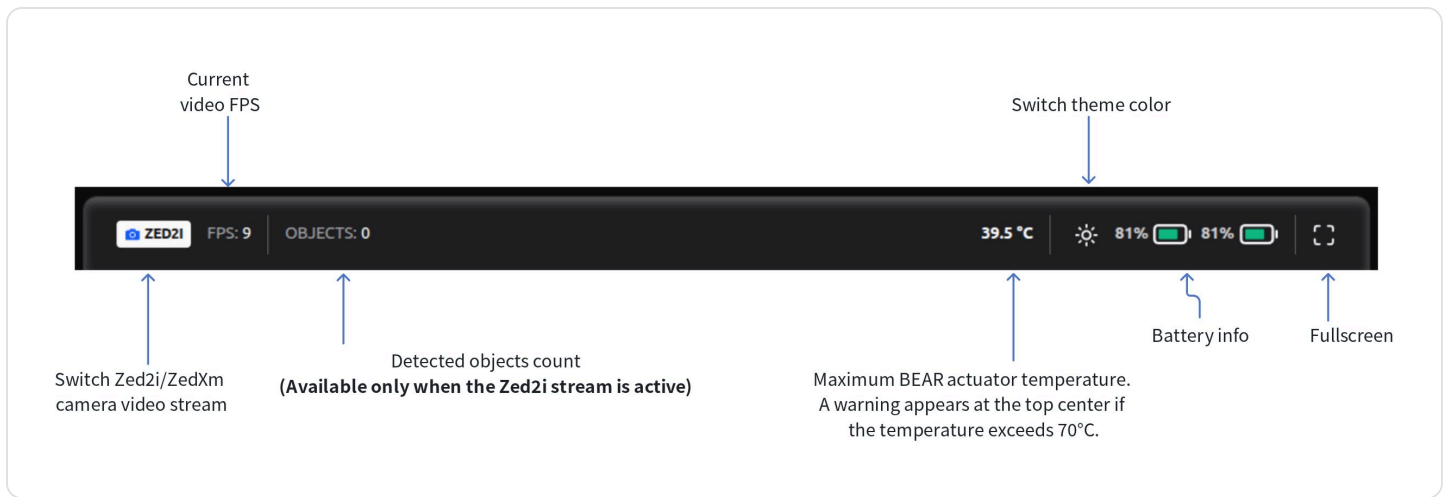
3.1 How to Access



As shown in the [THEMIS-UI](#) interaction diagram, the FPV page opens automatically after all startup steps on the Initialization page have completed successfully.

3.2 Page Content

- Top bar



Top Bar of the FPV Page

- The real-time video is streamed from either the ZED2i or ZEDxm camera. When the camera is switched, a loading prompt is displayed first and the view returns to normal once the new video stream becomes available.
- Colored detection boxes: objects automatically recognized by the system, each box is labeled with the detected object class. They are red by default, turn yellow when the mouse hovers over them, and turn green when selected or when the marker point falls inside the box.
- Yellow star: the current marker point.
- Blue star: the current interaction point.
- Purple direction line: the current grasp-axis direction.
- Path: the planned or travelled path of the robot.

3.3 Interaction Functions



FPV interaction functions are available only when the ZED2i video stream is selected.



If the high-precision recognition mode is enabled, the generic detection boxes whose name contains the string `object` are filtered out, and only the more specific categories are kept.

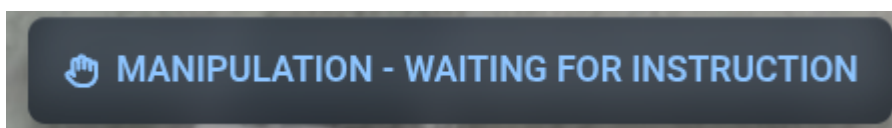
- Press and hold on the image for about 0.5 seconds (by holding the mouse button or keeping a finger pressed on the screen) to open the operation dialog:
 - Pressing and holding inside a detection box → the operation menu for that object; the title is the object class.
 - Holding on a blank area → the navigation and placement menu, titled "Navigation and Ungrasping".

- When the point is inside an object box, the options are:
 - Grasp the object in place — grasp directly without moving.
 - Navigate and grasp — move to that position first and then grasp (the coordinates are displayed).
 - Release the item at the target position — release / place the item at that target.
- When the point is on a blank area, the options are:
 - Move towards the position X, Y — navigate to the point that was clicked.
 - Place the item at X, Y — navigate to that point and place the item.
- After selecting one option, click Confirm to issue the command; clicking Cancel closes the dialog without performing any operation.



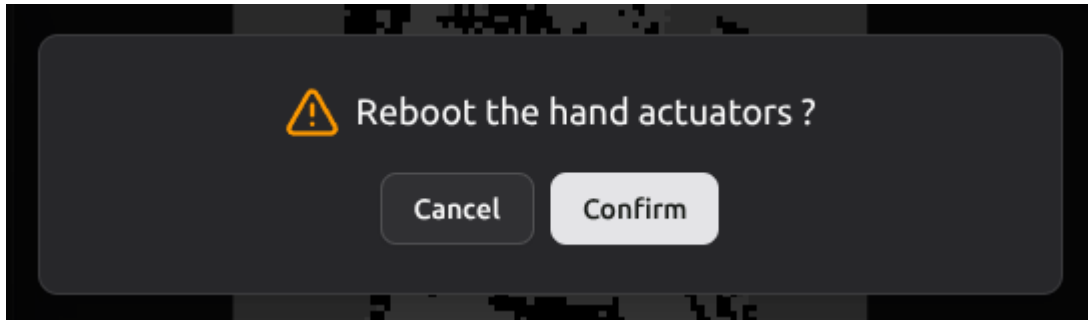
3.4 Grasp / Navigation Status

The blue message in the upper-right corner corresponds to the current manipulation or navigation task status, as shown below. It mainly includes:



- Waiting for command – standing by for the next operation
- Thinking – planning or making a decision (with a spinner animation)
- Grasping in progress / grasp complete
- Placing in progress / place complete
- Grasp finished
- Target unreachable – the current point is out of reach
- Task cancelled
- Greeting task complete

- Clicking to open a confirmation dialog for restarting the hand motors. Use this function if the fingers fail to move during grasping or exhibit abnormal motion:



4. Map Page

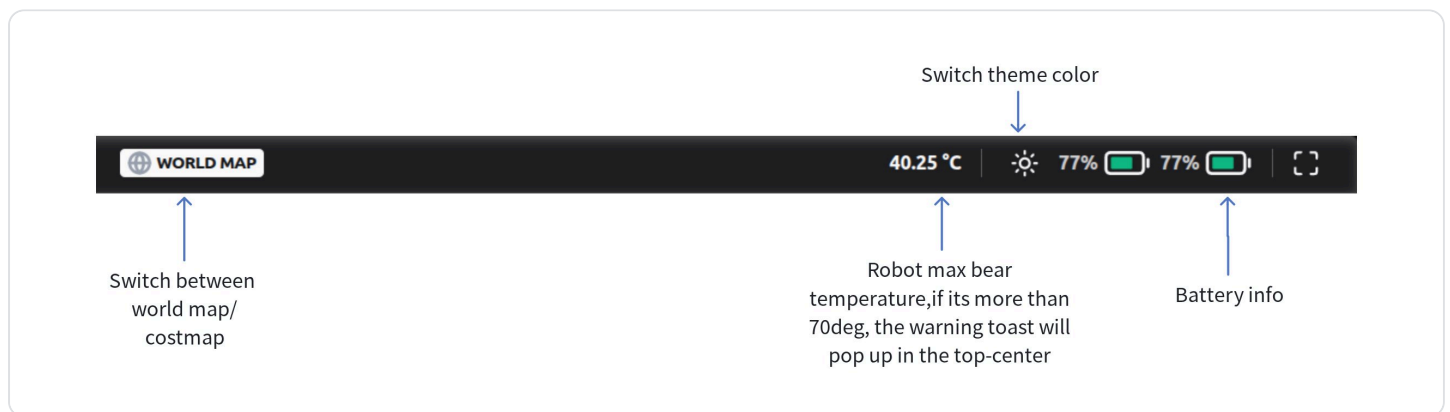
4.1 How to Access



As shown in the [THEMIS-UI](#) interaction diagram, open the Map page by selecting button B in the left navigation bar.

4.2 Page Content

- Top bar



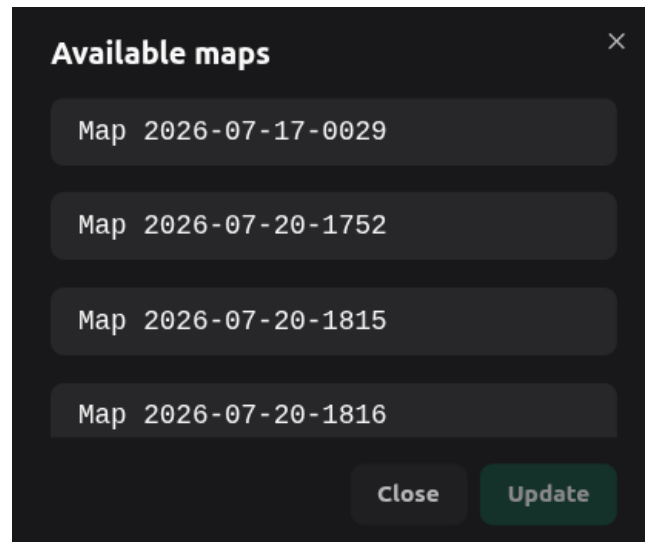
Top Bar of the Map Page

- World map: mainly shows the full environment map (PGM), which is the static map that has been loaded. If dynamic mapping is enabled, it is replaced by a map that is updated in real time.
 - Blue path line: the path currently planned by the robot
 - Robot icon: position + heading arrow
 - Destination pin: the navigation destination you have selected and confirmed
- Local map: mainly shows the local costmap around the robot (Costmap).

- Coloured cost grid — the real-time local map around the robot; the colour indicates the traversal cost
- Center arrow — the current heading of the robot
- Path line — the planned path overlaid on the cost map

4.3 Interaction Functions

- Globe icon in the lower-right corner: load a map. Click it to open the map selection dialog:
 - Select a map
 - Click “Update” to download and save it locally, and refresh the view at the same time

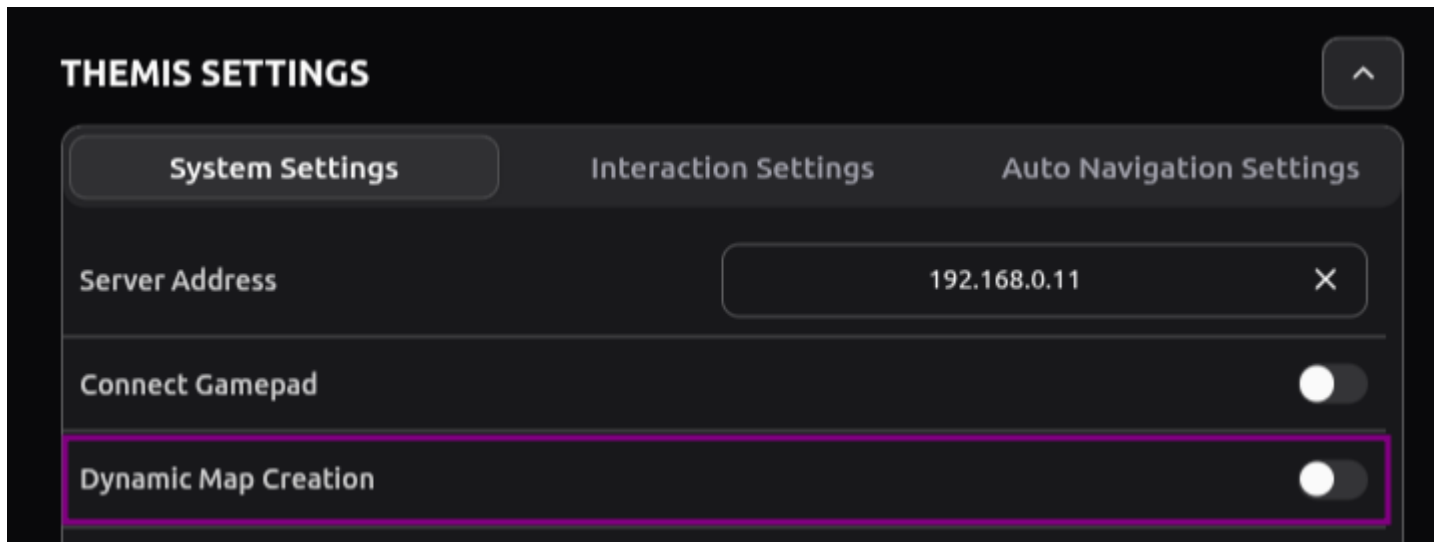


Load a Map



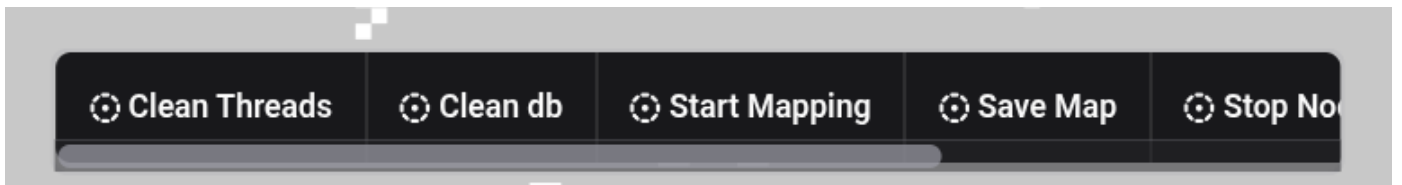
Use this function during initial setup or when switching to a different environment map. If the list fails to load, an error prompt is shown.

- Dynamic Mapping icon in the lower right: dynamic mapping (appears only after “Dynamic Mapping” has been enabled in the system settings; clicking it shows or hides the mapping progress bar at the bottom of the view).
 - After the **Dynamic Map Creation** mode is enabled, `Pgm2D` will no longer use the local static map but use the robot’s real-time mapping data as the base map.



Enable Dynamic Mapping

- Mapping progress: the operation flow panel in dynamic mapping mode arranges seven steps horizontally. Click a step to execute it. Only one step can run at a time. All steps are:
 - Clean Threads, Clean Database, Start Mapping, Save Map, Stop Node, Sync Map, Restart Localization



Steps of the Dynamic Map Creation

- The step being executed shows a spinner; a failure is marked in red with a prompt, and a success message is displayed when the step completes. It is suitable for completing “mapping while walking -> save -> sync -> relocalize” in order.

5. Third-person Page

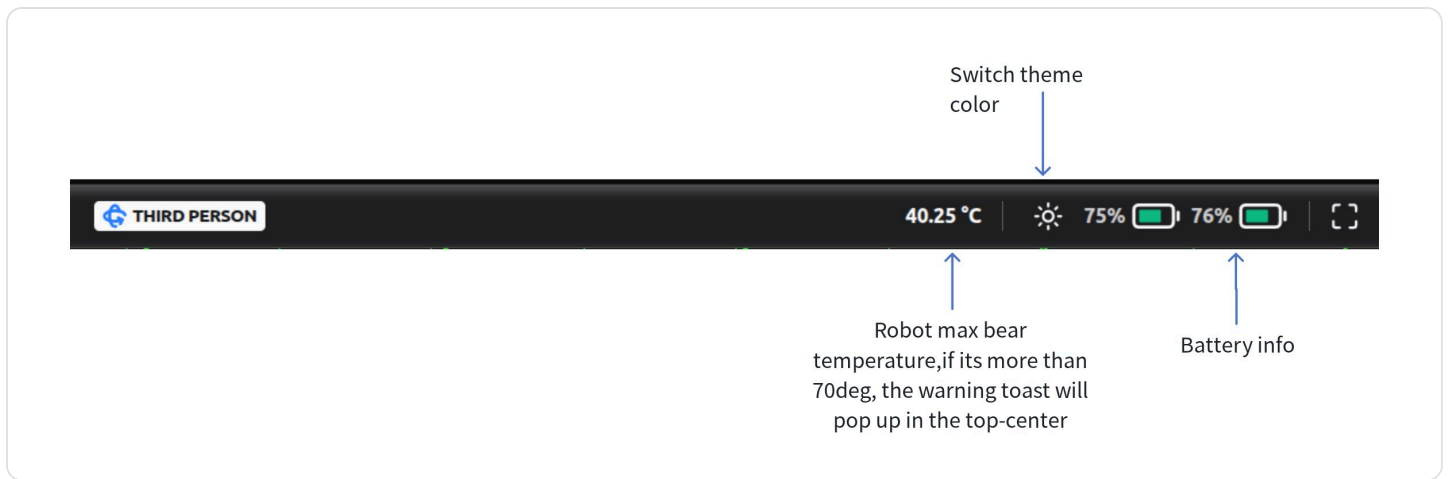
5.1 How to Access



As shown in the [THEMIS-UI](#) interaction diagram, this interface is entered by clicking the C button in the navigation bar on the left.

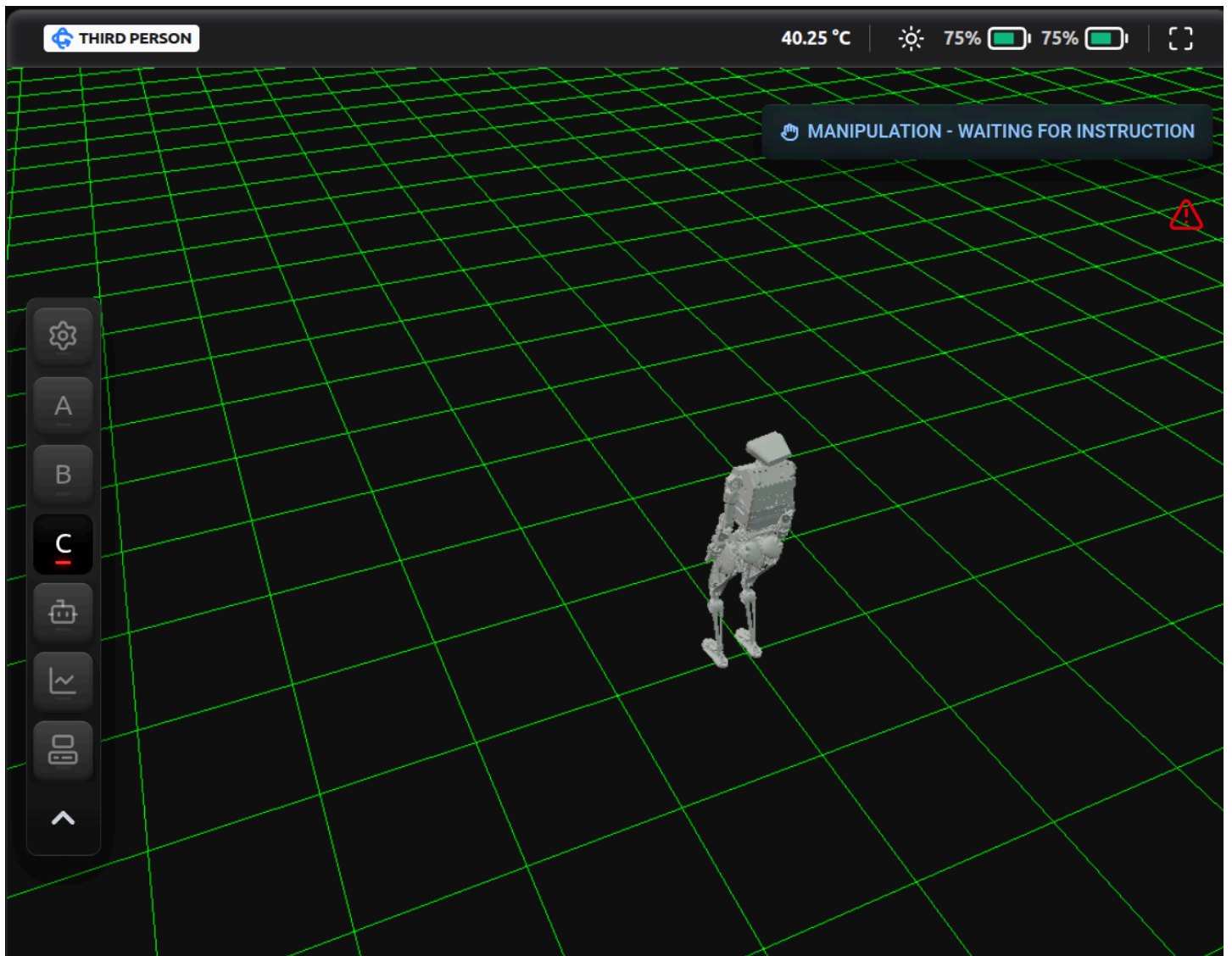
5.2 Page Content

- Top bar



Top Bar of the Third-person Page

- Loads the 3D model of the THEMIS robot and updates it with real-time data:
 - Joint angles -> the arms, body and other parts move accordingly
 - Body posture -> pitch, roll and yaw reflect the orientation of the robot
 - Robot position -> when the robot walks on the ground, the 3D model moves accordingly
- Hand motion trajectory
 - When hand trajectory data is enabled, colored points and trajectory lines are shown, which are used to observe the recent motion path of the arm or hand.



5.3 Interaction Functions

- Drag — rotate the view around the scene
- Scroll wheel — zoom in / zoom out
- Right-click and drag (or two-finger pan) — pan the view

6. THEMIS Settings Menu (Robot Settings)

6.1 How to Access

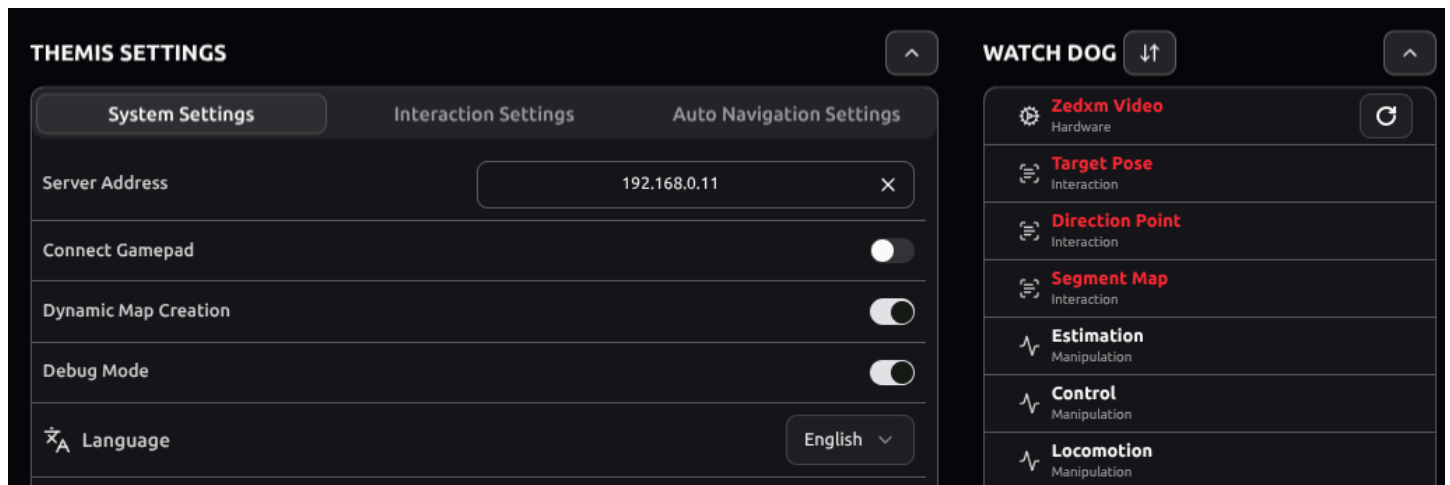


Open the [THEMIS Settings menu](#) by selecting the gear icon on Pages A, B, or C, or the gear icon in the upper-right corner of the Initialization page. The settings drawer opens from the bottom of the screen.

6.2 Page Content

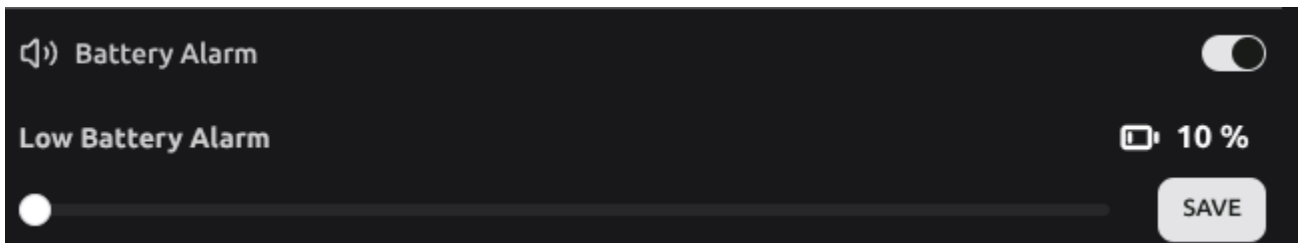


The settings drawer contains two panels. The left panel includes **System Settings**, **Interaction Settings**, and **Auto Navigation Settings**; the right panel displays **WatchDog**.



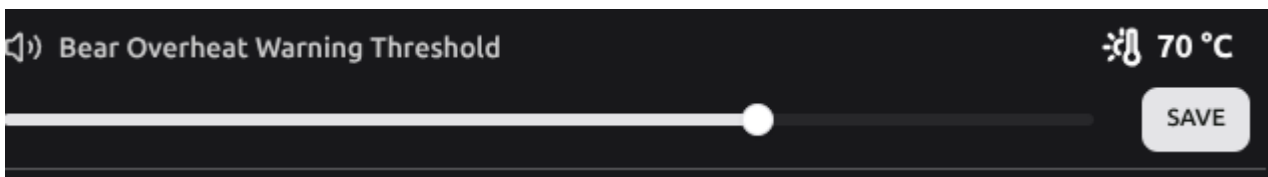
6.2.1 System Settings

- **Server address:**
 - Enter the IP of the robot and click Save; the format is then validated. If the address has changed, the real-time data and video connections are re-established and the page is refreshed after about half a second.
- **Connect gamepad:**
 - Switches gamepad input on or off.
- **Auto navigation:**
 - Enables or disables head tracking and autonomous-navigation behaviors through gamepad commands.
- **Dynamic mapping:**
 - When enabled, the [2D map page](#) uses real-time mapping data and displays the controls related to the mapping progress bar.
- **Debug mode:**
 - When enabled, the FPV and other pages display more performance debugging information.
- **Language:**
 - Switches the interface language between Chinese and English.
- **Battery voice alert:**



- When enabled, the battery level voice alarm threshold can be adjusted as required.
- **Note: Whether this option is enabled or not, the voice alarm is always triggered when the battery level is below 10%.**

- BEAR overheat temperature alert:



- If any monitored motor exceeds the configured temperature threshold, a voice alarm is triggered; the alarm threshold can be adjusted manually.
- The temperature alarm does not need to be switched on or off manually — it is always active.

- Semantic map:

- Switches the semantic segmentation map function; its state is linked with the `segmentMap` process in the watchdog.

- Grasp While Moving:

- Switches the “grasp while moving” capability by sending an enable or disable request to the server.

- Obstacle avoidance:

- Enables or disables obstacle avoidance through gamepad commands.

- Navigation task 1:

- A toggle that starts or stops the preset Navigation Task 1.

6.2.2 Interaction Settings



When switching to this tab, the current control parameters and detection model are loaded from the server. Selecting a different detection model allows the system to recognize different object categories.

- Detection model:

- Reads the detection models currently installed on the robot. Different detection models can be used to recognise different types of objects. For details, refer to the [THEMIS Deployment Manual](#).
- Detection types in use:
 - After one of the models is selected, the detection types already chosen under the current model are shown. For details, refer to the [THEMIS Deployment Manual](#).



The updated detection categories take effect in the real-time video on Page A. Objects that are not recognized will not be displayed with detection boxes..

- **Selectable Detection Types:**
 - After one of the models is selected, the detection types available under this model are shown. For details, refer to the [THEMIS Deployment Manual](#).
- **Confidence:** adjustable. The higher the value, the wider the detection range; the lower the value, the narrower the detection range.
- **Filter Generic Detections:** filters out the generic detection boxes whose name contains the string `object`, keeping only the more specific categories.
- **Grasp Speed** (0–100%)
- **Grasp Offset** (X / Y / Z, in cm, ± 40)

6.2.3 Auto Navigation Settings



All settings under this tab are closely integrated with the deployment demo workflows. For the deployment demonstration tasks, refer to the [THEMIS Deployment Manual](#), which contains a detailed description of all settings.

6.2.4 Process Monitoring (Watchdog)



Shows the running state of the robot background services and processes.

- Status colours — green = online and normal, red = offline / abnormal, grey / amber = other intermediate states
- The processes are divided into four categories: manipulation, hardware, navigation and interaction, specifically:
 - manipulation

- State estimation
- Control
- Locomotion
- Command processing
- hardware
 - Head computer online status
 - Chest computer online status
 - Body computer online status
 - Robot head process
 - Robot left / right leg processes
 - Robot left / right arm processes
 - Whether the head video stream is online
 - Whether the chest video stream is online
- navigation
 - Whether the costmap (costmap) is online
- interaction (interaction / perception)
 - targetPose — marker point (yellow star in FPV)
 - detectBox — object detection box (coloured box in FPV)
 - direction2D — 2D direction indicator (purple direction line in FPV)
 - segmentMap — costmap service status (the “Semantic Map” switch in the settings)
- Some processes have a restart button (rotating icon) on the right. Currently supported:
 - ZED2i (head camera video)
 - ZEDxm (front camera video)
 - Costmap (cost map)

6.2.4 Bottom Buttons



The bottom buttons are: **Restart Gamepad**, **Restart Video** and **Shutdown**. All three buttons will first open a confirmation dialog (with a warning icon) to prevent accidental operation.



7. Controller Devices Connection

7.1 Remote Control Devices

! With a mobile phone + game controller, some button combinations may activate built-in controller functions instead of sending commands to the robot. It is recommended to download the corresponding game controller app in advance, or to set up the controller button mapping beforehand.

7.1.1 Handheld Console, e.g. Steam Deck

You may choose to use popular handheld consoles available on the market to control THEMIS. We recommend using the Steam Deck for an optimal interaction experience and longer battery life. This document takes the Steam Deck as an example for explanation, though the button layouts and functions of other handheld consoles are very similar. Available options are as follows:

The functions of the handheld console buttons are as follows:

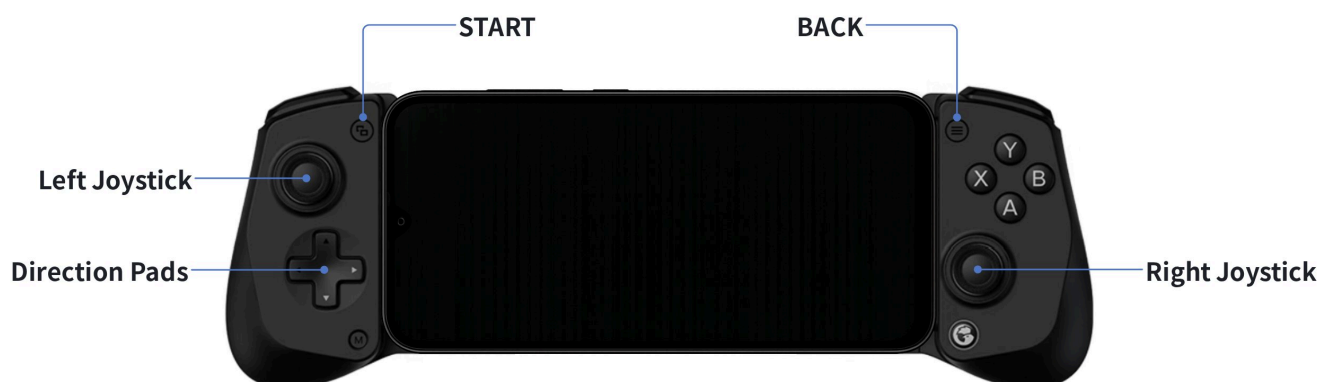


Steam Deck

7.1.2 Mobile Phone + Game Controller

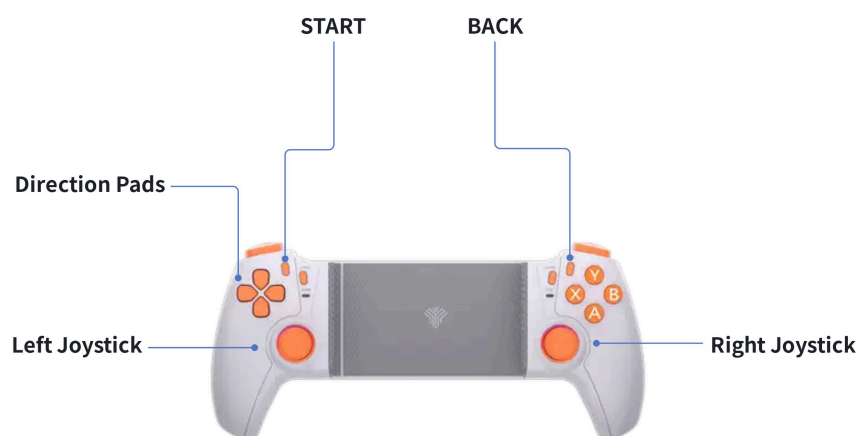
If a large display is not a priority, you may also choose to control THEMIS using a mobile phone paired with a game controller. We recommend the GameSir X5 Lite for a good user experience. This document uses the combination of GameSir X5 Lite with an Android

smartphone (Android 15) as an example, though the button layouts and functions of other controllers are very similar. The functions of the controller buttons are as follows:



x5 Lite

If you are using an iPhone, the Yuanqi-T1 game controller is recommended. It can be connected to your phone over Bluetooth, and its buttons for controlling the THEMIS robot are largely consistent with those of the GameSir X5 Lite.

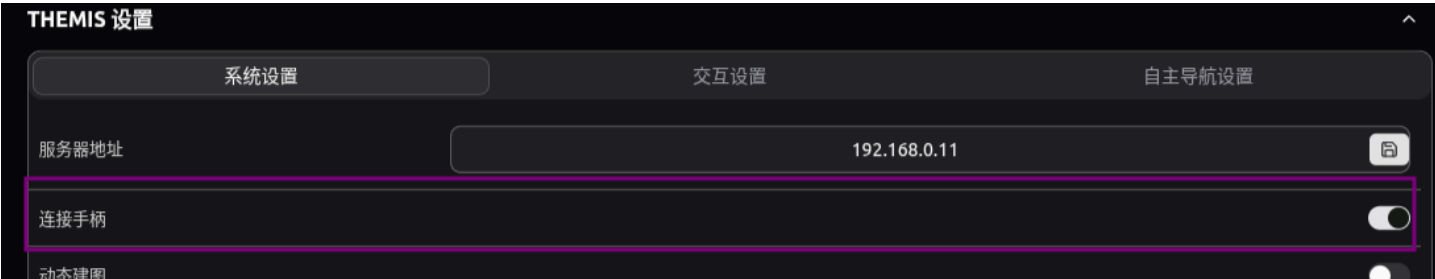


Yuanqi-T1

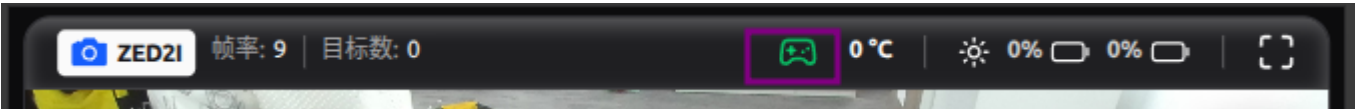
7.2 Connect / Disconnect the Controller

! Make sure the control device is connected to the robot' s onboard Wi-Fi and that the gamepad is connected to the control device.

First, turn on the Connect Gamepad switch in the THEMIS settings panel:



After turning it on, press the RT button once. If the controller is recognised, a connection-confirmation message appears, and a green controller icon is displayed in the top bar at the same time:



This means the controller is ready for use. To disconnect it, simply turn this option off; a “controller disconnected” message is displayed at the same time.

7.3 Remote Control Operation

Remote Controller Button Function Table	
Remote Control Input	Function
Mode Switching	
START + BACK	Standing Mode
START + Y	Walking Mode
Stand Mode	
Left Joystick X-axis (↔)	Center of Mass Lateral Shift↔
Left Joystick Y-axis (↑↓)	Center of Mass Vertical Offset↑↓
Right Joystick X-axis (↔)	Body Roll
Right Joystick Y-axis (↑↓)	Body Pitch
Shoulder Buttons LT/RT	Body Yaw Angle
Walking Mode	
Left Joystick X-axis (↔)	Linear Velocity: X-axis (Front/Rear)
Left Joystick Y-axis (↑↓)	Linear Velocity: Y-axis (Left/Right)

Shoulder Buttons LT/RT

Left/Right Turning